

emgforge: a first-principles-validated forward model of surface EMG, from volume-conductor lead fields to motor-unit action potentials and interference signals

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Abstract

Forward models of the electromyogram are used to explain motor-unit action potentials (MUAPs), to test decomposition algorithms on signals with known ground truth, and to train learning-based decoders — but they are usually validated against each other or against a handful of recordings, and the synthesis step that turns a lead field into a single-fibre action potential (SFAP) is rarely checked against the physics it claims to implement. We present **emgforge**, an open forward model with three things that are new. First, a synthesis method — the line-source integral with a monopole-fitted lead field, a second-derivative current source, a one-sided tendon window and physical time — that we show reproduces a closed-form line-source solution to $r = 1.0000$ and zero lag (with a monopole-free source), and that stays usable on finite-element lead fields because the monopole fit removes mesh ripple before the second derivative amplifies it. Second, a tiered validation suite for forward EMG models: fifty checks with numeric criteria drawn from the physiological literature, running from first principles through the analytical multilayer cylinder and the finite-element conductor to MUAP phenomenology (innervation-zone signature, end-of-fibre potential, depth decay, fat, electrode size, inter-electrode distance, comb filtering) and interference-EMG statistics. Third, the findings that suite produced: the spatial-frequency-domain engine that the field has used as a reference is anti-phase and lagged relative to the line-source integral and its propagating main lobe has inverted polarity; the width of a finite-element electrode source shapes the lateral footprint of the lead field; and on finite-element fields the recipe’s amplitude is erratic at $\pm 40\%$ while its shape and spectrum are faithful. The model also contains an anatomical tier — a segmented MRI forearm whose lead fields reproduce those distributed with the NeuroDec digital twin ($r = 0.999$ over 105 electrodes and 100 fibres), with harmonic-streamline fibres that stay inside the muscle by construction — and a motoneuron-pool and twitch layer that produces static and non-stationary interference EMG and force. Three datasets are released with the code: a cylinder SFAP atlas, a 100-unit forearm pool with its 5×5 HD-EMG MUAP tensor, and interference-EMG trials with ground-truth spikes.

1 Introduction

Forward models of the electromyogram simulate the signal an electrode records when muscle fibres fire. They are used to understand what shapes a motor-unit action potential, to test decomposition and estimation algorithms on signals whose ground truth is known, and, increasingly, to generate labelled training data for learning-based EMG decoding [1–3]. The classical models are analytical: a fibre in an infinite anisotropic medium [4–6], a planar layered conductor

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[7, 8], or a multilayer cylinder [9–11]. Finite-element models lift the geometric restrictions [12–14] and, with reciprocity and reusable source bases, can now be run on MRI-derived anatomy at interactive speed [1]. Generative surrogates trained on such twins [2, 3] make the output cheap and have been validated on their own terms (a 1.8% reconstruction error against the twin; movement EMG against recordings), but a surrogate inherits its twin’s synthesis step, and the finite-element twins themselves are not distributed — only the surrogates are. Open models exist at the planar level [15] and as multi-domain finite-element frameworks [16]; what has been missing is an open, anatomy-capable chain whose synthesis step is checked against the physics it implements rather than against another model.

Three of our own recent efforts depend on such a chain and motivated this one. MUniverse [17] benchmarks motor-unit decomposition on synthetic, hybrid and experimental HD-EMG and needs simulation stacks whose ground truth can be inspected; biophysical-model-informed decomposition [18] inverts a forward model to recover spike trains and unit properties, and is only as good as that model; and neural-operator surrogates of the lead field [19, 20] are trained on the finite-element fields described here and inherit the second-derivative sensitivity discussed in Sec. 4.

emgforge is an open, inspectable version of that chain, built on one rule: every step must be checkable in isolation — the lead field against an analytical model, the synthesis integral against a closed-form solution, the MUAP features against numbers from the physiological literature, and the whole chain against the statistics of real interference EMG. Applying that rule is where the new results of this paper come from. We write it plainly: what the model computes, in what order, with which parameters, and what happened when we tested it.

The contributions are:

1. **A synthesis method validated to first principles** (Sec. 4, 7.1). The *golden method* — the spatial line-source integral with a three-monopole fit of the lead field, a second-derivative current source, a one-sided tendon window and physical time — reproduces a closed-form line-source oracle at $r = 1.0000$ with zero lag and a monopole-free source. To our knowledge no published SFAP engine has been held to that test; the monopole fit is what makes the second-derivative integral usable on finite-element lead fields.
2. **A tiered validation suite for forward EMG models** (Sec. 7): fifty checks, each holding one measurable property of the chain against a literature principle with a numeric criterion, from first principles to interference statistics, with documented limitations kept as live criteria. The suite is reusable by any simulator that can produce a lead field.
3. **Findings.** The spatial-frequency-domain engine of Farina and Merletti [8], Farina et al. [11], which prior work (ours included) used as the shape reference, is anti-phase and lagged by 1–2.5 ms relative to the line-source integral and its propagating main lobe has inverted polarity; the width of a finite-element electrode source, not the conductor, sets the lateral footprint of the lead field; the recipe’s amplitude on finite-element fields is erratic at $\pm 40\%$ while shape and spectrum are faithful; and an electrode-placement artefact in an earlier layout had been misread as a model property.
4. **Anatomy from a mask:** harmonic-streamline fibre geometry (Sec. 5) that resolves contained, non-crossing fibres from a segmentation alone, on an MRI forearm tier whose lead fields reproduce those of the NeuroDec digital twin ($r = 0.999$, amplitude within 13%).
5. **A complete chain and its datasets:** three interchangeable volume conductors behind one lead-field interface (Sec. 2), a motoneuron-pool and twitch layer for static and non-stationary interference EMG and force (Sec. 6), and three released datasets with ground truth (Sec. 8).

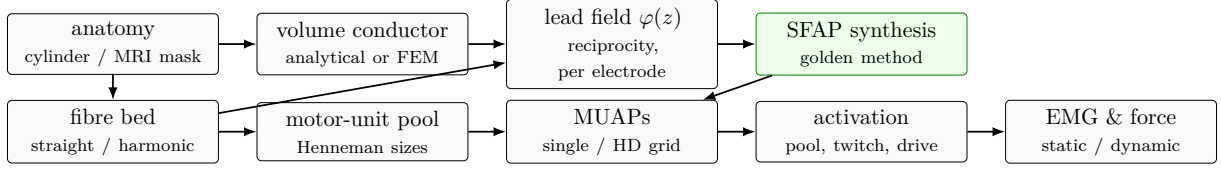


Figure 1: The `emgforge` chain. The lead field is the hinge: it is computed once per electrode and sampled along every fibre; the golden synthesis recipe turns each fibre’s $\varphi(z)$ into an SFAP; fibres are summed per motor unit, and the activation layer turns motor-unit spike trains into interference EMG and force.

2 Overview of the pipeline

Figure 1 shows the chain. Everything downstream of the volume conductor is pure NumPy; only the finite-element solve needs FEniCSx [21] and Gmsh [22].

Reciprocity. By the reciprocity theorem for a linear volume conductor [23], the potential at electrode e due to a unit current source at a fibre point $\mathbf{r}$ equals the potential at $\mathbf{r}$ due to a unit current injected at e . So one solve with the source at the electrode gives the *lead field* $\varphi_e(\mathbf{r})$ everywhere, and sampling it along a fibre path $\mathbf{r}(z)$ gives $\varphi_e(z)$, the weight with which membrane current at z reaches the electrode. Every fibre, every motor unit and every contraction reuse the same solve; only the number of electrodes costs finite-element work. The interior-point reciprocity of our solver is checked directly (Sec. 7.1).

3 Volume conductors and lead fields

3.1 Analytical multilayer cylinder

The reference conductor is the four-layer cylinder of Farina et al. [11] (bone, anisotropic muscle, fat, skin), solved in the spatial-frequency domain with Bessel functions; `emgforge` carries a line-by-line Python port of the original implementation (`emgforge.analytical`). We use it in two ways: its own signal generator produces MUAPs directly (the “Farina generator”), and its transfer function can be inverted to give the lead field $\varphi(z)$ along a fibre for a given electrode, which we feed to our own synthesis engine. The geometry and conductivities are in Table 1. The electrode is a disc of radius 5 mm; the coupled grid $\Delta z = v/f_s = 0.977$ mm (with $v = 4$ m/s, $f_s = 4096$ Hz) and 256 samples give a 250 mm span along the fibre.

3.2 Finite-element conductors

The FEM tier solves the quasi-static conductivity equation

$$\nabla \cdot (\boldsymbol{\sigma}(\mathbf{r}) \nabla u) = -s(\mathbf{r}), \quad (\boldsymbol{\sigma} \nabla u) \cdot \mathbf{n} = 0 \text{ on } \partial\Omega, \quad (1)$$

on a tetrahedral mesh with piecewise-constant tissue tensors (muscle transversely isotropic, ratio 5 along the fibre [24, 25]; fat, skin, bone isotropic). The source s is a zero-mean Gaussian blob of width σ_s centred on the electrode (its mean is subtracted so the pure-Neumann problem is solvable), and the linear system is solved with GMRES/ILU in FEniCSx [21]. Two families of mesh are built with Gmsh [22]: a *parametric limb* — layered discs (cancellous and cortical bone, muscle, fat, skin) extruded or lofted along z , circular or elliptical, optionally tapered, refined at the skin — and the *MRI forearm* (Sec. 3.3). For the cylinder used in validation the radii and conductivities are set equal to the analytical model’s (Table 1); the mesh has 1.03×10^6 cells and a solve takes ≈ 12 s. Figure 2 shows the resulting lead fields against the analytical ones. The width of the electrode source matters more than one might expect: with $\sigma_s = 5$ mm the blob

Table 1: Volume-conductor parameters. The validation cylinder uses the analytical values in both models; the production FEM/MRI table follows Gabriel et al. [26] at ≈ 100 Hz.

	bone	muscle $\sigma_{\perp}$	muscle $\sigma_{\parallel}$	fat	skin	radii (mm)
analytical [11]	0.02	0.10	0.50	0.04	1.0	10 / 35 / 38 / 40
FEM, validation cylinder	0.02	0.10	0.50	0.04	1.0	9, 10 / 35 / 38 / 40; $L = 240$
FEM / MRI, production	0.02 [†]	0.2455	1.228	0.0379	4.55×10^{-4}	from the segmentation

[†]cortical; cancellous bone 0.075 S/m. The skin values differ by 2000 $\times$ between the two tables: the analytical school treats skin as a thin conductive layer (1 S/m), the Gabriel-based tables as dry skin (4.6×10^{-4} S/m); it is the largest disagreement in the literature and a distinct thin skin layer changes the lateral decay [10].

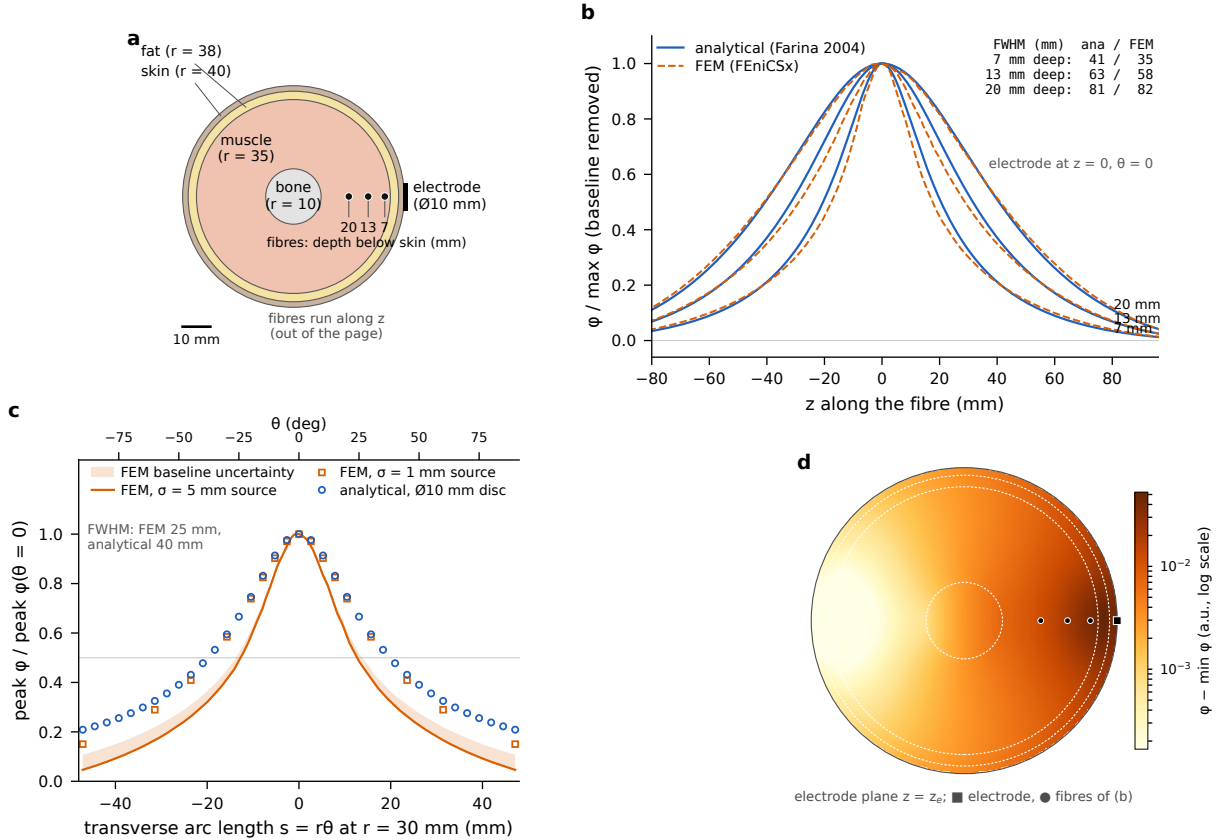


Figure 2: Lead fields on the validation cylinder. (a) Cross-section with the electrode and the fibre depths used. (b) $\varphi(z)$ along fibres 7, 13 and 20 mm below the skin: analytical (solid) and FEM (dashed); the widths agree for deep fibres and differ for the shallowest, where the electrode models differ (a $\varnothing 10$ mm disc against a Gaussian source that behaves as a $\varnothing 5$ mm disc). (c) Transverse decay at the fibre depth ($r = 30$ mm): the FEM field around the circumference (FWHM 25 mm; the band is the baseline uncertainty) against the analytical peak potential for fibres displaced around the circumference (FWHM 40 mm); with a $\sigma_s = 1$ mm source the FEM profile matches the analytical one (extra points), so the difference is the 5 mm electrode blob, not the conductor. (d) The FEM lead field of one electrode on the cross-section (log scale); the dots are the fibres of (b).

extends through the 2 mm skin and 3 mm fat into the muscle, and its transverse footprint at the fibre depth is narrower than the analytical one; a $\sigma_s = 1$ mm source reproduces the analytical transverse profile (Fig. 2c).

3.3 MRI forearm

A labelled segmentation (muscles, radius and ulna, fat/skin) is meshed into a tetrahedral forearm with a thin skin shell; each muscle’s conductivity tensor is rotated onto that muscle’s own centreline tangent, estimated from per-slice centroids and a smoothing spline (`sigma_mode="centerline"`). Skin electrodes are placed on the outer surface by ray-casting onto the mesh; the HD-EMG layout used in this paper is a regular 5×5 grid with 10 mm inter-electrode distance centred over the flexor carpi ulnaris (FCU), rows along the arm and columns around it (measured spacing 10.2 mm along, 10.0 mm across). The forearm mesh used here (subject WR, 42,315 cells) builds in 5 s and solves in 0.6 s per electrode; sampling φ along the 637 fibres of the FCU bed takes 6 s per electrode. A fresh solve of this model reproduces the lead fields distributed with the NeuroDec digital twin [1] for the same subject (Sec. 7.4).

4 The golden SFAP synthesis method

4.1 Line-source model

The action potential is the Rosenfalck form $V(s) = A s^3 e^{-s}$ for $s > 0$ (s in mm, $A = 96$ mV) [4], launched from the neuromuscular junction at z_0 in both directions and cut at the tendons by a window $w(z)$ over $[z_0 - L_1, z_0 + L_2]$:

$$V_m(z, t) = V(vt - |z - z_0|), \quad V_m^w(z, t) = V_m(z, t) w(z), \quad (2)$$

so that $t = 0$ is the instant the junction fires and the wave front reaches the proximal tendon at L_1/v and the distal one at L_2/v . A fibre of radius a and intracellular conductivity σ_{in} injects, per unit length, the membrane current of core-conductor theory [4, 27],

$$i_m(z, t) = \sigma_{\text{in}} \pi a^2 \frac{\partial^2 V_m^w}{\partial z^2}, \quad (3)$$

and the potential at the electrode is the lead-field-weighted sum of that current,

$$\text{SFAP}(t) = \int i_m(z, t) \varphi(z) dz = \sigma_{\text{in}} \pi a^2 \int V_m(z, t) w(z) \varphi''(z) dz, \quad (4)$$

where the second form follows by integrating by parts twice (the boundary terms vanish because V_m^w has compact support) and shows that the SFAP is the inner product of the windowed action potential with the *second derivative* of the lead field. Taking $\partial^2/\partial z^2$ of the whole windowed field, rather than of each travelling half analytically, is what produces the three source terms that the literature separates by hand [28, 29]: the two propagating tripoles, the *generation* term at the cusp $|z - z_0|$, and the *end-of-fibre* terms where the window cuts the wave. These are exactly the terms that make the source monopole-free, $\int i_m dz = 0$ for all t [15]; we verify this to 8×10^{-17} (Sec. 7.1).

Units and constants. φ from the finite-element tier is the potential, in volts, for a unit current (1 A) injected at the electrode against a uniform volume sink (the zero-mean source of Sec. 3; the potential is therefore defined up to a constant, which φ'' removes). The analytical φ is recovered from the Farina transfer function up to an arbitrary constant, so the analytical tier is used for shape and relative laws and absolute amplitudes are quoted only from the finite-element tier. The engine uses $\sigma_{\text{in}} = 1$ S/m and $a = 50 \mu\text{m}$ (a $100 \mu\text{m}$ fibre, at the large end of the $40\text{--}80 \mu\text{m}$ physiological range; amplitude scales with a^2), so the constant $\sigma_{\text{in}} \pi a^2$ is a scale factor rather than a calibrated quantity and absolute microvolt values carry that caveat.

Table 2: The golden recipe (`SpatialConfig`). The cylinder tier uses $f_s = 4096$ Hz; the MRI tier 2048 Hz with $v = 3.3\text{--}4$ m/s.

parameter	value	why
<code>denoise</code>	monopole, 3 poles	fit the analytic form of φ ; removes mesh ripple before φ''
<code>edge_taper_left/right</code>	5 / 10 samples	Gibbs guard at the sampling-window ends
<code>upsample_factor</code>	2	second derivative off the Nyquist zigzag
<code>csd_derivative</code>	2	physical current source (triphasic)
<code>fiber_window</code>	one-sided, $\alpha = 0.25$	flat at the junction, tapered at the tendons
<code>center_time / t_start_ms</code>	false / -10	physical time, junction at $t = 0$
v, f_s, w	4 m/s, 4096 Hz, 256	coupled grid $\Delta z = v/f_s$
σ_{in}, a	1, 0.05 mm	amplitude constant $\sigma_{\text{in}}\pi a^2$; no $1/v$ (the IAP is fixed in space)

4.2 The recipe

Discretised, Eq. (4) is one matrix–vector product, $\text{SFAP} = (\mathbf{C}\varphi)\Delta z$, with $\mathbf{C}$ the $(n_t \times n_z)$ current-source-density matrix. The steps around it, and the reason for each, are (Fig. 3, Table 2):

1. **Monopole denoise of φ .** A finite-element $\varphi(z)$ carries mesh-scale ripple. Because the SFAP integrates φ'' , that ripple is amplified quadratically in spatial frequency. Instead of low-passing — which rounds the real peak and, as we show in Sec. 7.2, halves the spectral content — we replace φ by a least-squares fit of the form the field actually has, $\hat{\varphi}(z) = \sum_{i=1}^3 A_i / \sqrt{d_i^2 + (z - z_i)^2} + c$, a sum of three free-position point sources. On a clean analytical φ the fit is a near no-op ($r > 0.99$); on a FEM φ it changes the field itself by at most 1.8% of its peak but lifts the correlation of φ'' with the analytical one from 0.47 to 0.91, cuts the ripple in φ'' eighteen-fold, and takes the SFAP’s jaggedness from 0.011 to 0.002 (the analytical value is 0.0018).
2. **Edge taper.** A short cosine ramp (5 and 10 samples) on the two ends of the sampled φ , a Gibbs guard for lead fields truncated by the sampling window.
3. **Upsampling $\times 2$** (cubic spline). The second derivative needs a finer grid than the raw φ or the SFAP develops a Nyquist zigzag; $\times 2$ removes it and $\times 4$ adds nothing.
4. **Current-source density.** $\mathbf{C}$ is built from the full bidirectional windowed field of Eq. (2) by two numerical z -derivatives.
5. **One-sided tendon window.** The window $w(z)$ is flat through the junction and cosine-tapered over the last 25% of each semi-fibre only. A symmetric Tukey window would notch the junction, where the wave is born; a hard boxcar produces a sharp end-of-fibre spike (physically right for one fibre with a sharp tendon, and the choice we use when comparing to sharp-tendon oracles) which in real muscle is softened by tendon scatter. The taper costs $r = 0.998$ against the sharp-tendon oracle.
6. **Physical time.** No centring: $t = 0$ is the junction, the detector lobe of an electrode Δz from the junction lands at $\Delta z/v$, the end-of-fibre onset at L/v ; a 10 ms pre-roll is prepended so a junction under the electrode is not pinned to the window edge.

4.3 The spatial-frequency engine and our port of the Farina generator

`emgforge` also carries a spatial-frequency-domain engine after Farina and Merletti [8], Farina et al. [11], in which φ is transformed along z , multiplied by the fibre-end function, the action-potential spectrum and jk_z , and brought back to time by a Radon section, together with a

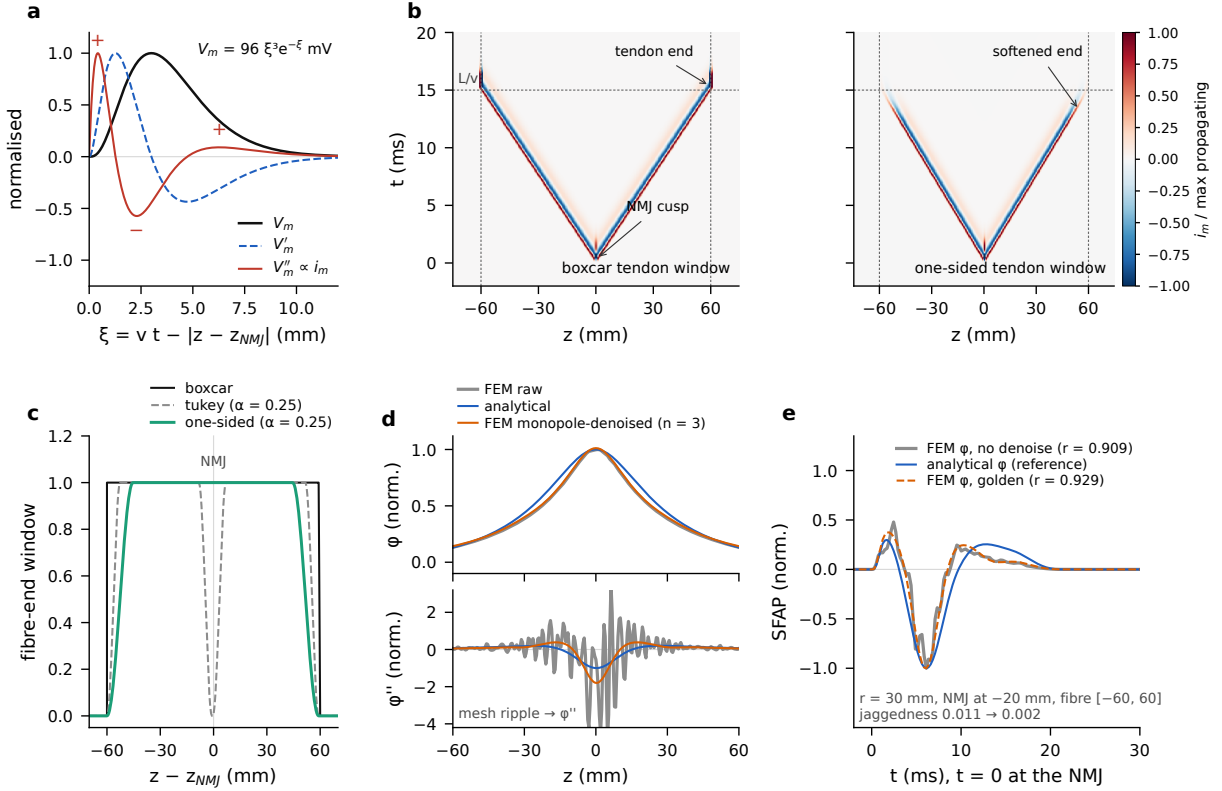


Figure 3: Anatomy of the golden method. (a) The Rosenfalck action potential and its first and second spatial derivatives; the second derivative is the triphasic current source. (b) The current-source-density matrix $C(z, t)$ for a fibre $[-60, 60]$ mm with the junction at 0: two tripoles leave the junction, the cusp at the junction and the window edges at the tendons appear as sources; boxcar (left) and one-sided (right) tendon windows. (c) The three fibre-end windows. (d) A finite-element lead field raw, monopole-denoised and analytical, with their second derivatives (normalised to the analytical peak; the colour scale in (b) is clipped at the propagating maximum — the tendon term is $2.7\times$ larger). (e) The SFAPs each produces: the ripple in φ'' reaches the raw SFAP and the monopole fit removes it (r against the analytical- φ SFAP $0.909 \rightarrow 0.929$); the remaining difference from the analytical waveform is the FEM field’s own shape at this shallow depth (Sec. 7.1), not ripple.

line-by-line port of the Farina generator itself (Sec. 3). The engine was our first production engine and the reference the spatial engine was originally tuned against. The validation in Sec. 7.1 shows that, given the same φ , the engine does not reproduce the line-source integral — it is anti-phase and lagged by 0.9–2.6 ms — while its amplitude agrees with the spatial engine’s to within 5%; and that in the ported generator the end-of-fibre onset leads L/v by 2.5 ms and the propagating main lobe has the opposite sign to the textbook convention. We attribute these to the orientation and sign conventions of the action potential in our port; the original model was verified against limiting cases and reproduced by finite elements to 3–5% [1], and a convention difference in a port is not an error of the model. Until that audit is done we treat the spatial recipe as the reference and keep the other two for comparison.

4.4 From SFAPs to a MUAP

A *fibre bed* object carries, per fibre, the sampling step, the two semi-lengths, the junction position and the conduction velocity; a MUAP is the sum of the SFAPs of the fibres in a motor unit, each computed from its own $\varphi(z)$ with the shared configuration. Junction scatter, conduction-velocity scatter and tendon scatter are therefore explicit per-fibre values rather than filters: a 5 mm junction scatter across 50 fibres lengthens the MUAP from 20.0 to 22.2 ms and reduces its peak-to-peak to 0.86 of the coherent sum (Sec. 7.2), the temporal dispersion real MUAPs

show. Because time is physical, this dispersion is real and not a window artefact — the reason the spatial engine, and not the Fourier one, carries production.

5 Fibre geometry and motor units

5.1 Straight fibre beds

For a parametric limb, and as the default for the MRI forearm, fibres are placed on the muscle’s cross-section at mid-length by Poisson-disk sampling at a chosen density (4 fibres/mm² for the FCU, which oversamples the pool: the 100-unit pool only needs a subset) and extended as full-length paths parallel to the muscle centreline, with one global semi-length and a junction at an atlas innervation-zone fraction (0.305 of the length for the FCU). This is the classical assumption; it is simple, and it is what the literature’s cylinder models assume. Its cost is discussed in Sec. 9.

5.2 Harmonic-streamline fibres

Straight, constant-depth, full-length fibres are a poor description of a real muscle belly. N. Ezaz-Nikpay contributed a method that resolves a muscle’s internal architecture from its label mask alone: solve the masked Laplace problem $\nabla^2\phi = 0$ inside the muscle with $\phi = 0$ on the proximal cap, $\phi = 1$ on the distal cap and an insulated surface, and trace the streamlines of $\nabla\phi$ (second-order Runge–Kutta) from seeds on the mid-belly cross-section. Streamlines are contained in the muscle and never cross by construction. In the production model each streamline is one full-length fibre with a single junction placed at the muscle’s shared innervation zone; a quality guard rejects tracer failures (turn per step above 15°, or a longitudinal span below 30% of the muscle). An experimental variant cuts each streamline into short in-series fibres, each with its own junction at an atlas innervation band — the series-fibering that long muscles show [30]; it narrows the surface field to real motor-unit widths but is not yet validated, and is gated behind a flag. On the FCU, seeded densely enough to match the straight bed (0.5 mm seed grid; the library default of 2 mm gives only 52 streamlines), the harmonic bed has 827 fibres that stay inside the mask for 99.96% of their length against 96.6% for the 637 straight fibres; their junctions lie within 1.2 mm (s.d.) of the shared innervation zone. The streamlines are not all full length: the median spans 0.85 of the muscle but a quarter span less than 0.53 (the straight bed spans 0.89 throughout), a bimodal distribution that the guard keeps because short edge fibres may be real; deciding that needs anatomical data. With the junction at that zone the two beds give near-identical MUAPs ($\approx 0.26 \mu\text{V}$ for a matched unit), so the harmonic advantage at this stage is anatomical rather than electrical (Fig. 4). One convention to watch: the harmonic model measures its junction fraction along the muscle’s principal axis, which for the FCU points the other way from the straight bed’s fibre coordinate, so the same number places the junction at opposite ends; Fig. 4 therefore innervates both beds on the same plane. The released datasets use the straight bed.

5.3 Motor-unit pool

Motor units are drawn from the bed following the size principle [31]: N sizes log-uniform between 5 and 400 fibres (many small units, few large), each unit anchored on a random bed fibre and its territory radius grown until it contains the target number of fibres, territories allowed to overlap; units are sorted by size so that index i is recruitment order. Because the bed is a fixed sample of the muscle (637 fibres for 11,207 fibre slots in the 100-unit pool), fibres are shared between units — 18 units per fibre on average — which is a coarse but consistent sampling: a fibre’s SFAP is computed once and reused. A unit’s MUAP is the sum of its fibres’ SFAPs; Fig. 5 shows the pool and its MUAPs. The amplitude–size relation in Fig. 5d is therefore a

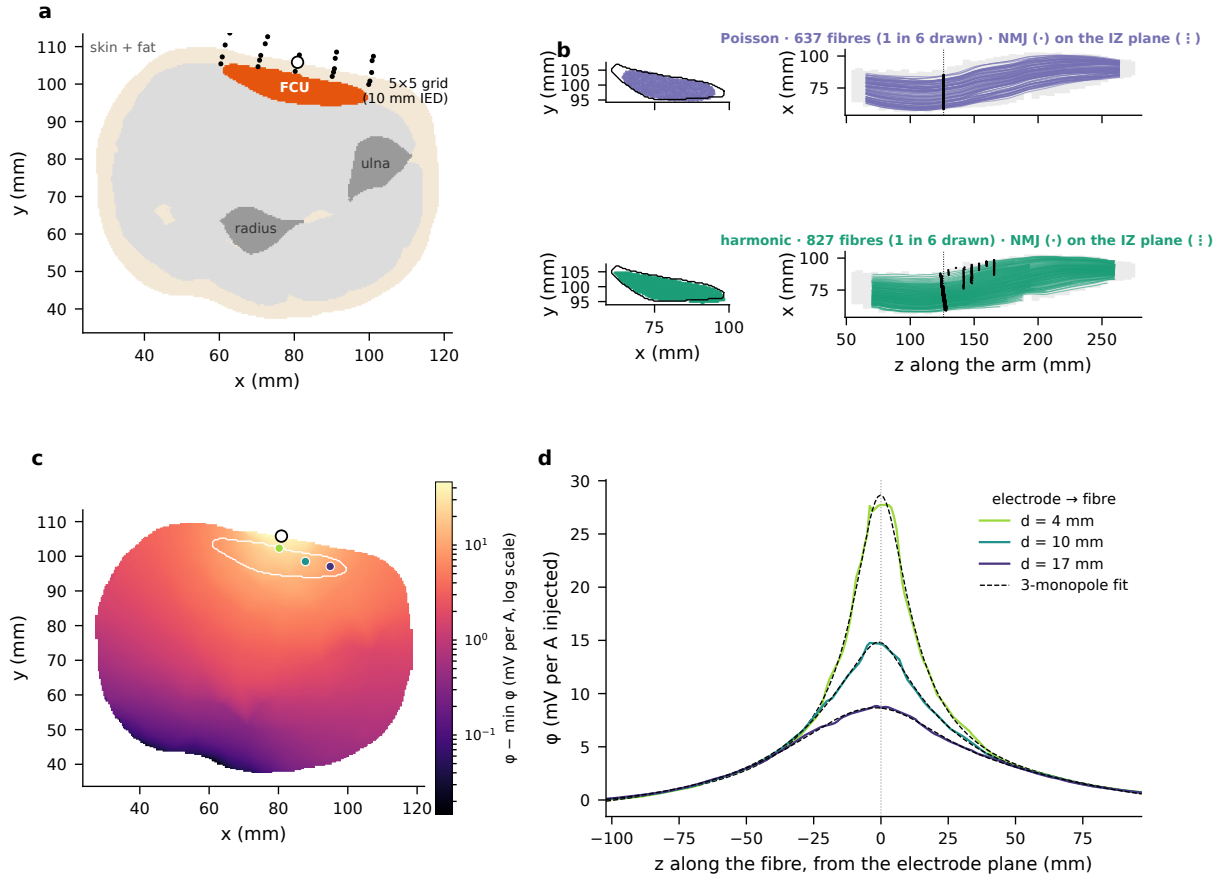


Figure 4: The MRI forearm (subject WR). (a) A segmentation slice with the FCU highlighted and the 5×5 electrode grid projected. (b) The FCU fibre bed at 4 fibres/mm^2 : 637 straight Poisson fibres and 827 harmonic streamlines (one in six drawn), both innervated on the same plane (dotted; the atlas fraction 0.305 of the straight fibres' length): 84% of the streamlines cross it and are innervated within 1 mm of it; the rest start on its far side ($z = 140\text{--}165 \text{ mm}$) and are innervated at their nearer end, the second cluster of dots. (c) The reciprocal lead field of one grid electrode on a mesh slice, $\varphi - \min \varphi$ on a log scale (as in Fig. 2d). (d) φ along three fibres 4, 10 and 17 mm from the electrode, in mV per ampere injected, with the three-monopole fit the golden recipe uses (dashed).

consistency check (with shared fibres at similar depths, amplitude must scale with fibre count) rather than evidence.

6 Activation and interference EMG

The activation layer is phenomenological, written from the equations of the Fuglevand pool as parametrised in NeuroMotion [3, 31]. Given a scalar excitation $E(t) \in [0, 1]$:

- *Recruitment*: unit i has a threshold RTE_i spread exponentially with range 50 up to $E = 0.75$ for the last unit.
- *Rate coding*: once recruited, the rate is $\min(\text{PFR}_i, \text{MFR}_i + (E - \text{RTE}_i) s_i)$ with the peak rate falling from 40 to 30 pps, the minimum from 10 to 5 pps and the slope from 50 to 30 pps per unit drive across the pool — the onion skin [32].
- *Spikes*: a renewal process with Gaussian inter-spike intervals of coefficient of variation $1/6$, re-drawn at each discharge; units de-recruit when E drops below threshold. A common-drive fluctuation (low-passed noise at 2 Hz) can be added to E .

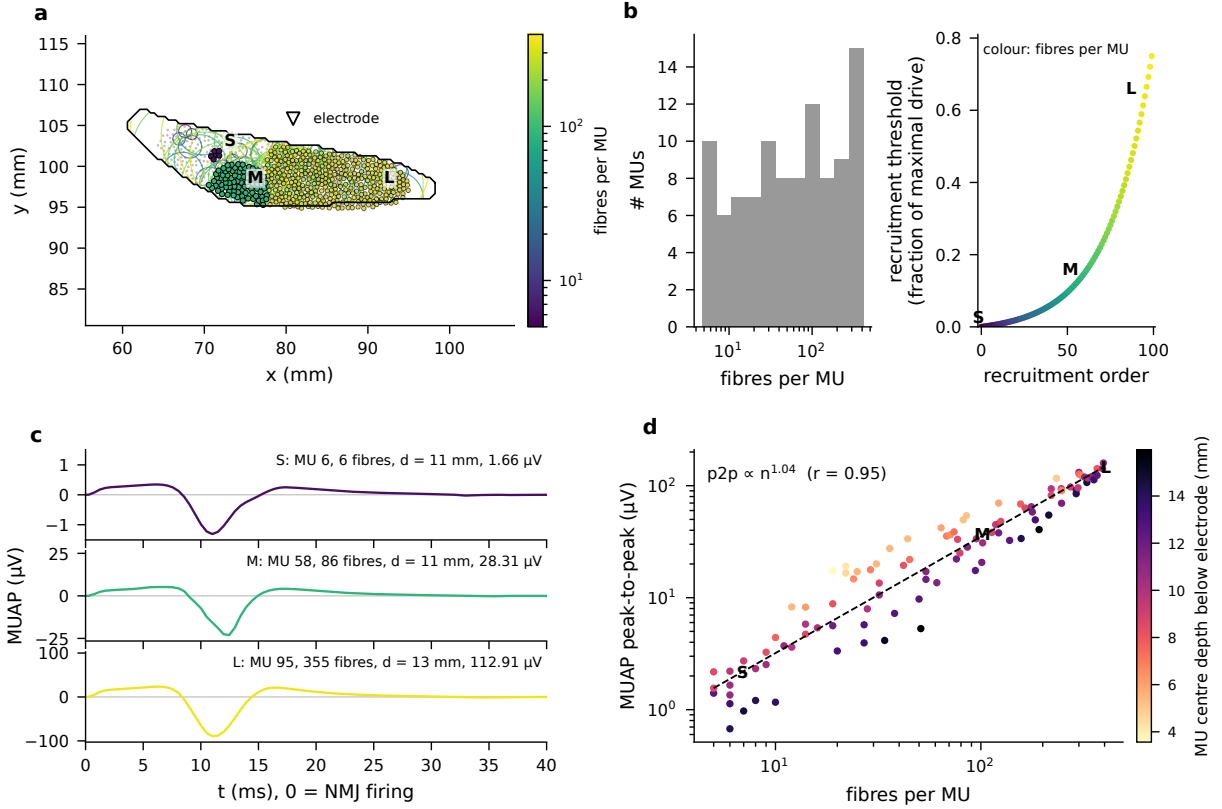


Figure 5: The 100-unit FCU pool. (a) Territories on the cross-section, colour by size (5–395 fibres, median 66; territory radii 0.6–13.8 mm), with the centre electrode of the grid. (b) Size distribution, and recruitment threshold against recruitment order (colour: size). (c) Golden MUAPs at the centre electrode of a small, a medium and a large unit (1.7, 28 and 113 μV ; across the pool 0.7–160 μV , median 24, median duration 18 ms). (d) MUAP amplitude against fibre count: $p2p \propto n^{1.04}$ ($r = 0.95$), the scatter being depth (colour); with shared fibres this is a consistency check (Sec. 5).

- *Force*: each discharge adds a twitch $P_i(t/T_i) e^{1-t/T_i}$ with amplitudes spanning 100 $\times$ and larger units contracting faster, scaled by the rate-dependent fusion gain of Fuglevand et al. [31]; the sum is normalised to the mean force at full excitation, so force is in %MVC.
- *EMG*: each spike places the unit’s MUAP (single channel or on every electrode of the grid) and the trains sum; single- and double-differential montages are differences of electrodes.
- *Non-stationary EMG*: a normalised joint angle $\theta(t)$ sets the drive ($0.08 + 0.42\theta$), scales each spike’s MUAP ($1 + g_a(\theta - 0.5)$; shortening brings fibres to the electrode) and time-warps it ($1 + g_w(\theta - 0.5)$), with library defaults $g_a = 0.7$, $g_w = 0.18$ (the trial in Fig. 6e uses 0.8 and 0.22). This is a modulation layer, not a musculoskeletal model; models of the geometry change itself exist [33, 34] and are the next step.

Figure 6 shows a unit on the grid, an interference plateau, the pool’s recruitment and rate coding, a trapezoid contraction with its force, and a movement trial.

7 Validation

The suite (`scripts/validation/`, ≈ 4 min) holds one measurable property of the chain per check against a literature principle with a numeric criterion, in four tiers: (A) the cylinder must reproduce — first principles, then the analytical model, then the finite-element model, then the pipeline on each; (B) the MUAP phenomenology the literature agrees on; (C) the statistics of interference EMG and the pool; (S) the chain-level sanity checks that ship with the simulator.

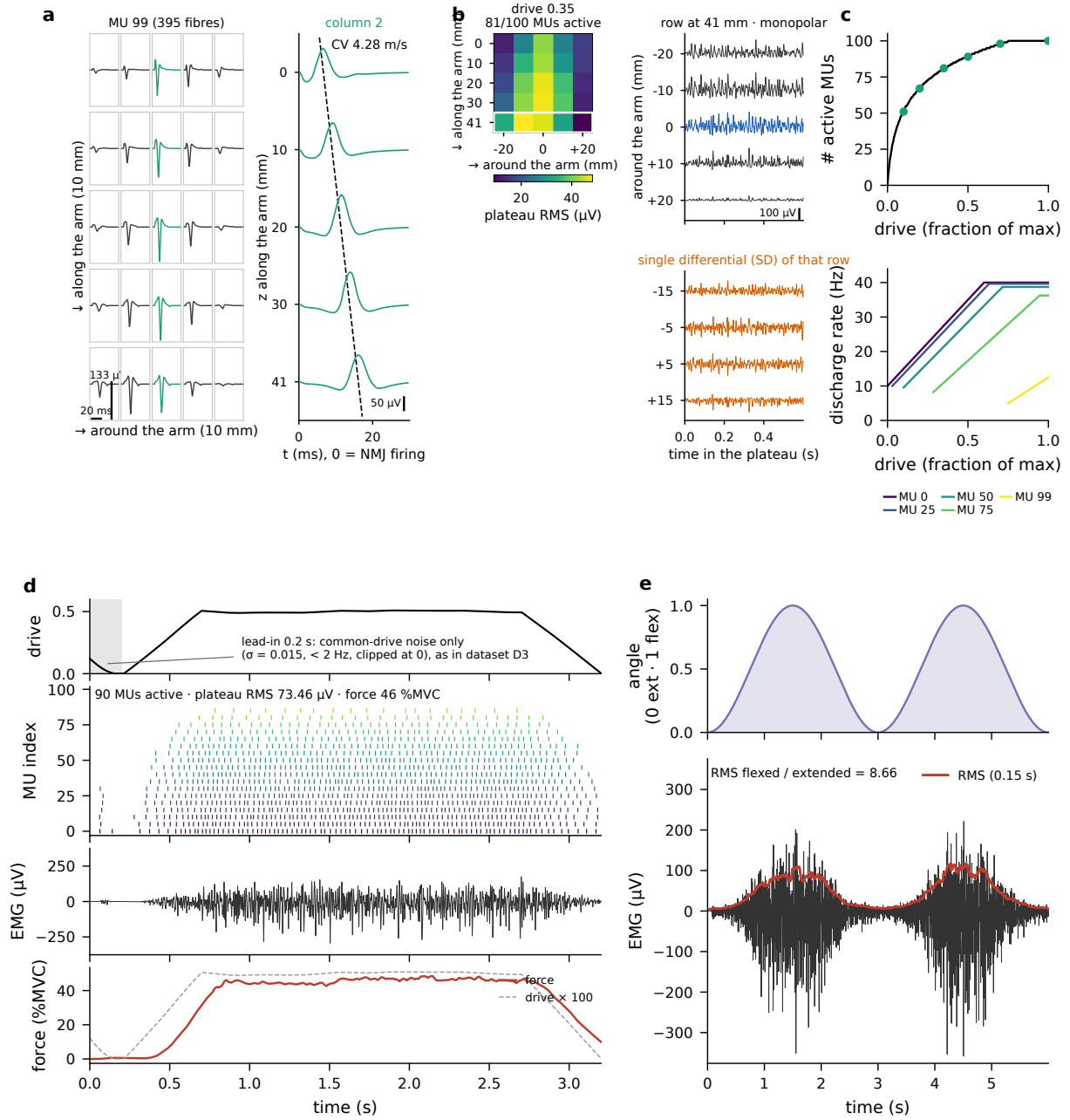


Figure 6: From motor units to signals (regular 5×5 grid, 10 mm spacing). (a) The largest unit's MUAP on the grid — rows run along the arm, columns around it — and, right, the column with the largest amplitude as a waterfall offset by electrode position: the wave walks down the column at 4.28 m/s (dashed guide; 4.1–4.8 over the pool, set 4.0). (b) Interference HD-EMG at a 35%-drive plateau (81 of 100 units active): the RMS map of the grid ($9\text{--}48\ \mu\text{V}$), one row of monopolar traces and the single-differential traces of that row at a shared scale. (c) Recruitment and rate coding of the pool. (d) A trapezoid contraction at half drive: drive (the lead-in carries the common-drive fluctuation), spikes, EMG (plateau RMS $73\ \mu\text{V}$), force (46% MVC); across drives 0.1–1.0 the plateau RMS is $14\text{--}133\ \mu\text{V}$ and the force 7–100% MVC. (e) A movement trial: angle and non-stationary EMG (RMS flexed/extended 8.7).

Table 3: Validation scoreboard (branch `paper/arxiv-v1`; tiers C and S run on the released forearm pool, dataset D2). “pass / total” counts every check; “known” are documented limitations that fail without failing the tier; the two genuine failures are discussed in Sec. 9. Appendix A lists all 50 checks.

tier	scope	pass / total	known	fail
A	cylinder: first principles, analytical, FEM, pipeline	16 / 20	3	A1.1b: FEM φ'' at mid-depth, $r = 0.93$
B	MUAP features vs literature	14 / 14	0	—
C	interference EMG and pool	6 / 8	1	C2: no refractory floor
S	chain-level sanity	8 / 8	0	—

Table 3 is the scoreboard, Appendix A lists every check with its criterion and measured value, and the full report is in the repository (`docs/validation/`). A “known” flag marks a documented limitation that keeps its criterion, so it turns green when fixed, without failing the run; “pass / total” counts every check. The prose below gives the three results that matter — the engine equals the integral, the spatial-frequency family does not, and the finite-element amplitude and depth law are open — and the phenomenology as a list of measured facts.

7.1 Tier A: the cylinder must reproduce

First principles. A fibre in an infinite anisotropic medium has a closed-form lead field, $\varphi(z) \propto (\rho^2/\sigma_\rho + z^2/\sigma_z)^{-1/2}$ [25, 27], whose second derivative is analytic. Evaluating Eq. (4) in the dual form on a 0.02 mm grid gives an oracle with no numerical derivative of the tendon step and no volume-conductor model in the loop. The spatial engine, given the same φ on its 0.977 mm grid, matches it with $r = 1.0000, 0.9999, 0.9999$ and lags 0.00, $-0.05, -0.05$ ms for junctions 0, $-20, -30$ mm from the electrode (Fig. 7a). This is a test of the discretisation of the integral with the same action potential — the physics of the integral is settled — and it is what makes the later comparisons attributable. The source is monopole-free to 7.7×10^{-17} ; the SFAP is invariant to the sampling rate and step ($r > 0.998$, amplitude within 0.3%); moving the electrode by 20 mm delays the propagating lobe by 4.88 ms ($\Delta z/v = 5.00$, one sample); superposition is exact; and changing v from 4 to 3 m/s stretches the SFAP by 1.328 (4/3) and scales its mean frequency by 0.757 (3/4) [35]. The oracle also caught a defect: the first version of the engine divided by v once too often (a constant inherited from the spatial-frequency formulation, where the $1/v$ belongs), so its ratio to the oracle was exactly $1/v$ — 0.497 at $v = 2$, 0.249 at $v = 4$. With the division removed the ratio is 0.994 at every v from 2 to 5 m/s (Fig. 7c); the residual 0.6% is the discretisation of the 0.977 mm grid. The correction also dissolved a long-standing puzzle in the repository: the spatial and Fourier engines had been pinned as disagreeing in amplitude by a “non-constant” 3–5 $\times$; that was the $1/v$ factor across the velocities of the reference set, and the two engines’ amplitudes now agree to 0.95 ± 0.05 , independent of v . The Fourier engine on the same φ scores $r = -0.79, -0.68, -0.64$ at lags of 1.4, 2.6, 0.9 ms (Fig. 7b).

Analytical against finite-element lead fields. Along fibres 7–20 mm below the skin the FEM $\varphi(z)$ correlates with the analytical one at $r = 0.995$ –1.000, and at $r = 0.997$ –1.000 for fibres 10–45° off the electrode meridian. Their widths differ for the shallowest fibres (FWHM 41 vs 35 mm at 7 mm), and comparing the second derivatives against analytical electrodes of different radii shows the FEM Gaussian source behaves as a 5 mm-diameter disc; at mid-depth (13 mm) φ'' agrees only to $r = 0.93$ whatever electrode is assumed — the one tier-A failure. The pipeline run on FEM φ against the pipeline run on analytical φ gives $r = 0.90$ –0.97 for shallow fibres and 0.998 for deep ones; the Fourier route, which low-passes φ at ≈ 32 mm wavelength, gives 0.995–1.000 at all depths, which is how earlier reports of “ $r \approx 0.99$ ” were obtained.

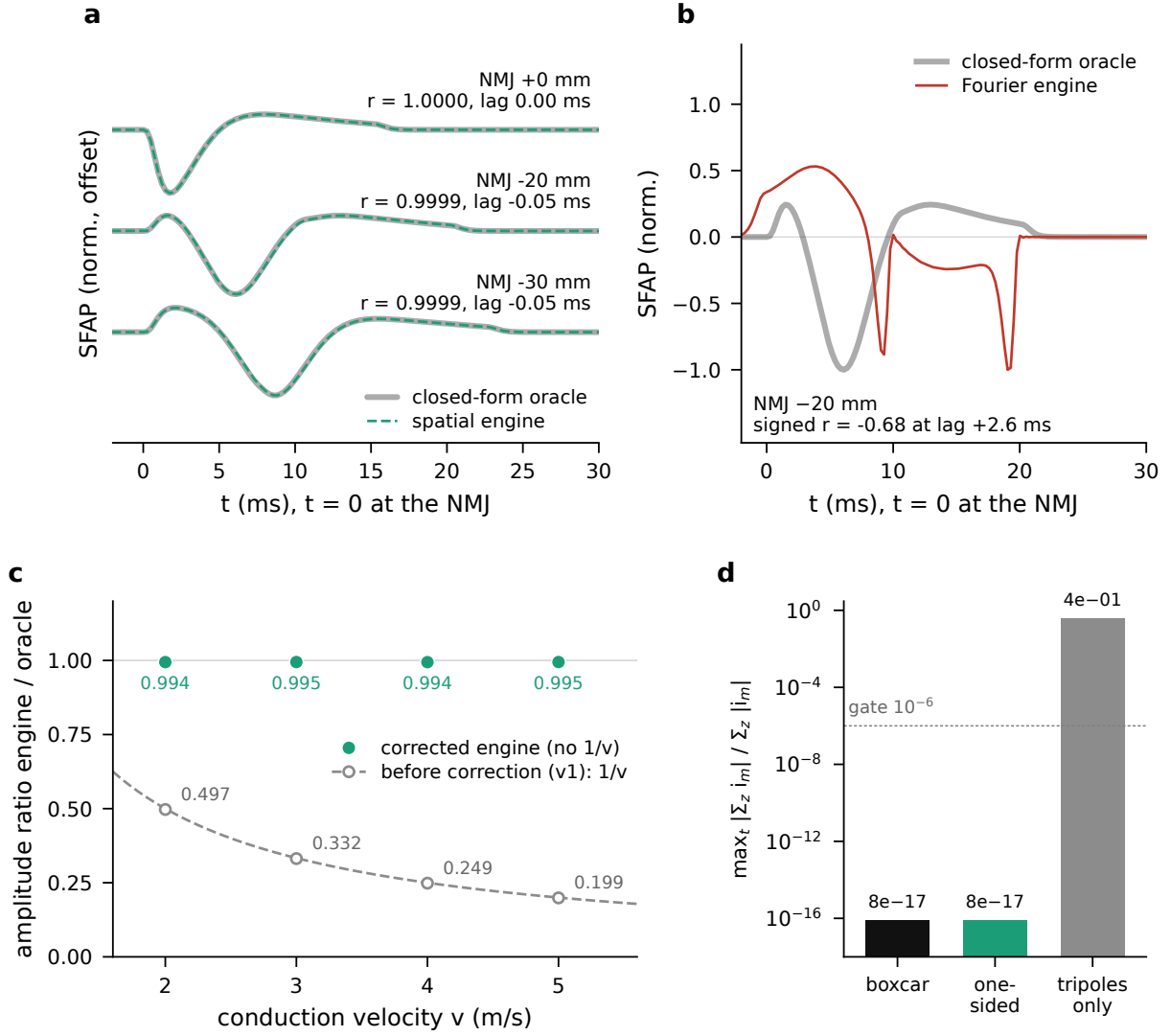


Figure 7: First principles. (a) The closed-form line-source SFAP and the spatial engine, three junction positions. (b) The Fourier engine against the same oracle. (c) The engine-to-oracle amplitude ratio: 0.994–0.995 at every v after the correction described in the text; the dashed $1/v$ is what the uncorrected engine gave. (d) The net source current at every instant is zero to machine precision once the junction and end terms are included; the propagating tripoles alone leave 39%.

Timing and symmetries. Isolating the end-of-fibre component by subtracting the same fibre with the tendon moved 60 mm further, its onset lands at 10.26 and 15.39 ms for $L/v = 10$ and 15 in the pipeline (within 0.4 ms; 0.6 ms across a nine-electrode array, where the threshold crossing depends on the local waveform); in the Farina generator it leads by up to 2.5 ms. Conduction velocity from a four-electrode array is 3.98 m/s (pipeline) and 3.83 m/s (Farina) for $v = 4$. Rotating the electrode by 30° equals rotating the fibre ($r = 1.000$, peak within 1%); shifting it 20 mm along the axis equals shifting the sampling window ($r = 0.996$ – 0.999 , the residual being the 240 mm mesh’s end effect [12]); and the lead field between two interior points is reciprocal to 1.35%.

7.2 Tier B: MUAP features against the literature

Each check is run on the Farina generator (the field’s reference model) and on our pipeline, with arrays and montages built by translating the lead field (Fig. 9, Fig. 8):

- *Morphology:* between the junction and the tendon the monopolar SFAP is + − + with

the main lobe negative [36]. The pipeline is; the Farina generator’s main propagating lobe is positive.

- *Conduction velocity* from a single-differential array recovers 3.13, 3.98, 4.88 m/s for set values 3, 4, 5 [37].
- *Innervation zone*: monopolar potentials mirror about the junction ($r > 0.998$); the single-differential channel over it is 4% of the maximum and the channels either side are mirror images of opposite sign ($r < -0.998$) [38].
- *End of fibre*: the same onset on every electrode (0.00 ms spread); its ratio to the propagating part grows monotonically with depth (0.02 at 7 mm to 0.20 at 20 mm) and is suppressed mono > SD > DD (0.053, 0.014, 0.004) [39, 40].
- *Depth*: peak-to-peak amplitude is a power law in depth over 7–25 mm ($R^2 > 0.98$), exponent 2.9 monopolar and 3.5 single-differential [30, 41].
- *Fat*: RMS retained at 3/9/18 mm of fat is 0.50/0.23/0.12 against the finite-element curve 0.69/0.20/0.10 of Kuiken et al. [42]; mean frequency falls; the FEM tier attenuates like the analytical one within 30% once φ is smoothed.
- *Electrode and montage*: amplitude and mean frequency fall with electrode radius; a 5 mm disc is -2.8 dB at 100 cycles/m [36]; single-differential amplitude doubles from IED 2.5 to 5 mm and saturates beyond 10 mm; with a single travelling wave, the two electrodes of a bipolar pair see shift-copies to 2.2×10^{-3} (1.4×10^{-3} without the monopole fit), so the montage is the textbook comb filter with nulls at $n v / \text{IED}$ [35, 43].
- *Anisotropy* elongates the lead field along the fibres by exactly $\sqrt{5}$ in the infinite medium and by 1.5 in the layered cylinder.
- *Linearity*: 50 identical fibres give $50\times$ the SFAP to 10^{-15} .

7.3 Tier C and S: interference EMG and the chain

These tiers run on the released forearm pool (Sec. 8). The pool’s rates lie in 5–40 pps with the onion-skin ordering ($\rho = -0.98$ between threshold and rate); inter-spike-interval variability is 0.16; thresholds are right-skewed and twitches span $95\times$; the amplitude distribution is super-Gaussian at low drive (kurtosis 5.6 at 5%) and Gaussian at high (2.9–3.1 at 50–90%), with ARV/RMS 0.77–0.79, between the Laplacian and Gaussian values [44, 45]; the RMS at half maximal force is 0.56 of that at maximal force, inside the linear-to-quadratic band of Lawrence and De Luca [46]; the median frequency at moderate drive is 77 Hz (mean 81 Hz); larger units give larger MUAPs ($\rho = 0.98$). The chain-level checks pass: orderly recruitment, onion skin, EMG rising with drive, force monotone and calibrated to 100% MVC at full drive, conduction velocity 4.33 m/s read off the HD grid (4.1–4.9 over all 100 units, 10 mm inter-electrode distance) for a set 4.0 m/s — the 8–24% overestimate is the apparent velocity $v / \cos \theta$ of fibres running $5\text{--}25^\circ$ to the grid columns (the same estimator recovers 3.98 on the cylinder, and inclination bias of this size is documented [47]); the check’s criterion is the physiological band — a localised MUAP footprint (peak/mean RMS 2.2), an $8.7\times$ amplitude modulation across a movement, and a MUAP scale inside the physiological band (median peak-to-peak $23.6 \mu\text{V}$, range $0.7\text{--}160 \mu\text{V}$, median duration 18 ms, against tens to hundreds of microvolts and $5\text{--}20$ ms [36]). Amplitude cancellation is flagged (68% already at 20% drive against $\approx 33\%$ in Keenan et al. [48]); see Sec. 9.

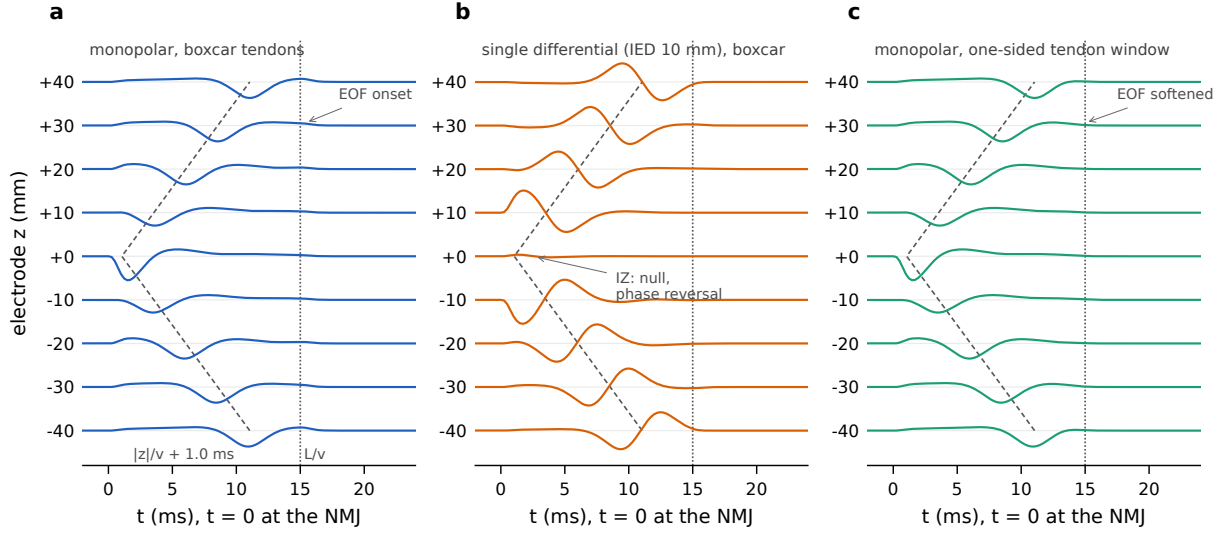


Figure 8: Physical time on a finite-element lead field (fibre 10 mm deep, tendons at ± 60 mm, $v = 4$ m/s). (a) Monopolar SFAPs along a nine-electrode array with hard tendons: the propagating lobe walks at $|z|/v$ (a fit gives 4.04 m/s; the guide is drawn at $|z|/v + 1.0$ ms because the negative phase trails the wave front by the action potential’s rise) while the end-of-fibre potential starts at 15.4–15.6 ms on every channel ($L/v = 15$ ms; spread 0.24 ms). (b) The single-differential montage vanishes and reverses at the innervation zone. (c) The same array with the one-sided tendon window of the golden recipe: the end-of-fibre potential falls from 0.16 to 0.10 of the propagating amplitude.

7.4 MRI tier

The NeuroDec digital twin [1] distributed, for subject WR, the reciprocal lead field of each of its 105 skin electrodes sampled along 100 fibre paths. Re-solving the same mesh with our solver and sampling the same paths (`scripts/validation/mri_neurodec_check.py`; `artefact docs/validation/neurodec_wr.json`) gives a signed correlation of 0.9994 on average (0.9989–0.9999 over the 105 electrodes) and an amplitude ratio of 1.13 (median; 1.03–1.23), i.e. the same field up to a 13% scale that we attribute to the source normalisation and skin shell of the two solvers — the volume conductor is the same physics on the same anatomy. On that anatomy the harmonic fibres are contained in the muscle mask for 99.96% of their length (96.6% for straight fibres). A mesh refinement study of the 42,315-cell forearm mesh against a finer one has not been done and is listed in Sec. 9.

8 Released datasets

Three datasets are generated by scripts in `paper/figures/` and described by manifests in `paper/datasets/` (Table 4). They are meant as reference material: every array can be re-generated from the repository.

9 Limitations and open items

We list these in the order we would fix them.

1. **Conduction velocity and amplitude.** With the constant corrected, the engine’s amplitude is independent of v because its action potential is fixed in space. Physiologically the action potential is fixed in *time* and a faster fibre carries a longer wave, which is why single-fibre amplitude falls with conduction velocity [6]; reproducing that requires stretching the action potential in space with v , a modelling choice we have not yet made.

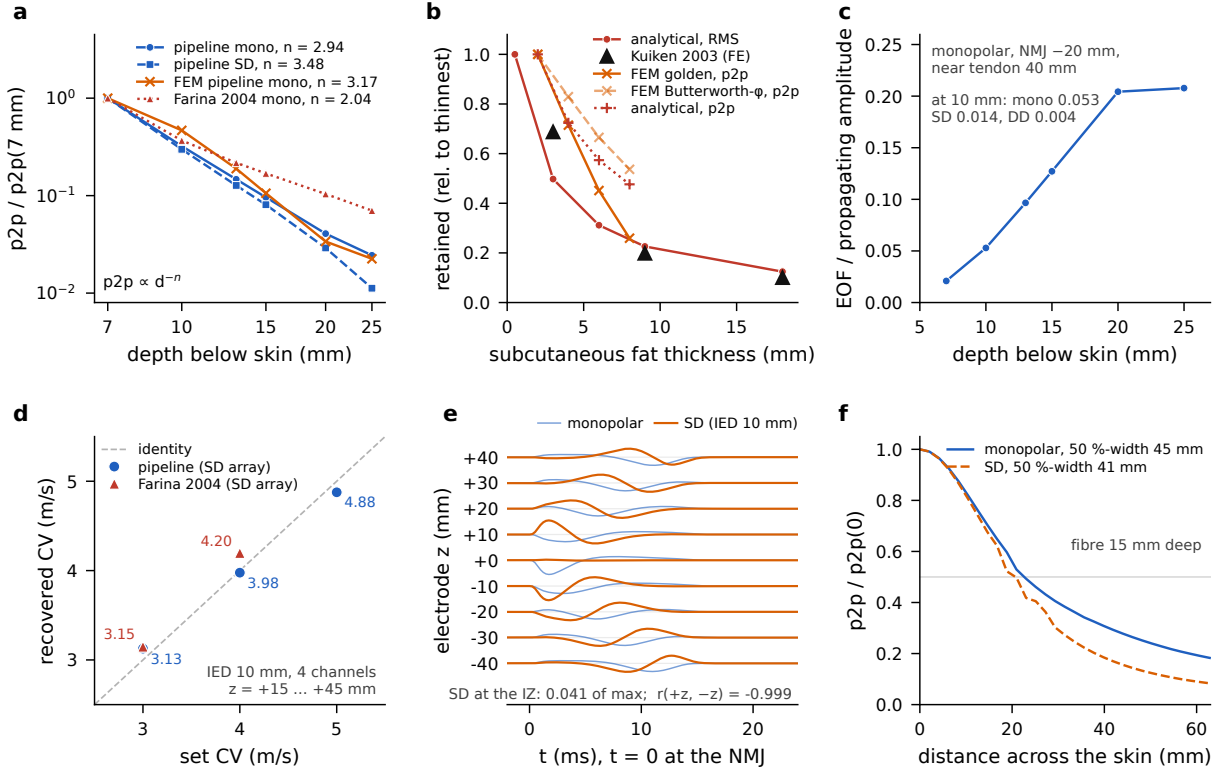


Figure 9: MUAP features against the literature (fibre 10 mm below the skin unless stated; $v = 4$ m/s). (a) Depth law of the peak-to-peak amplitude (log-log) for the pipeline (monopolar and single-differential), the Farina generator and the FEM pipeline, with fitted exponents. (b) RMS retained against subcutaneous fat thickness, analytical model against the finite-element points of Kuiken et al. [42]; FEM pipeline (golden and Butterworth-smoothed) and Farina generator at 2–8 mm, each relative to its thinnest case. (c) End-of-fibre to propagating amplitude ratio against depth (junction 20 mm and near tendon 40 mm from the electrode). (d) Conduction velocity recovered from a four-channel single-differential array against the set value. (e) The monopolar and single-differential array across the innervation zone (IED 10 mm). (f) Transverse spread of monopolar and single-differential amplitude for a fibre 15 mm deep.

Table 4: Datasets released with the paper (float32, NumPy .npz with a JSON manifest each; 8 MB in total).

name	contents	size
cylinder_sfap_atlas	analytical and FEM lead fields and golden SFAPs on the validation cylinder for 6 depths $\times$ 5 angles $\times$ 3 junction offsets $\times$ 9 electrode positions (1620 SFAPs), with the Farina-generator MUAP of each case	1.6 MB
forearm_fcu_mu_pool	the 100-unit FCU pool (fibre paths, fibre indices, sizes, territories, activation parameters), golden MUAPs at the centre electrode and the full (100, 25, 256) HD-grid MUAP tensor, electrode positions, configuration	2.4 MB
interference_emg_trials	trapezoid contractions at six drive levels (0.1–1.0): drive, spike trains, single-channel and 25-channel EMG, force; one movement trial	4.1 MB

2. **Fourier engine.** Given a lead field, it is anti-phase and lagged relative to the line-source integral, and its end-of-fibre onset is early. We suspect the orientation and sign conventions of the action potential in the Radon formulation ($V2 = -\text{flip}(\dots)$); until audited it should not be used as an oracle.

3. **FEM lead-field fidelity for the recipe.** On finite-element φ the golden SFAP amplitude

is erratic at $\pm 40\%$ across depth: the recipe integrates φ'' faithfully down to ≈ 8 mm wavelengths, where FEM φ carries mesh structure that the three-pole fit does not remove (five or seven poles do not either). Low-passing φ makes amplitudes smooth but halves the SFAP’s mean frequency (46 vs 112 Hz golden, 91 Hz analytical). A finer mesh around the fibre band, or a smoothing-free local estimate of φ'' , is the fix. Until then FEM amplitude laws should be read through the smoothed route; shape, timing and spectrum can be read from the golden one.

4. **Depth law.** The FEM cylinder decays $\approx 30\%$ slower with depth than the analytical one over 7–20 mm, whatever the preprocessing or the source width. This is a volume-conductor difference still to be attributed (mesh resolution near the skin against the analytical model’s spectral truncation). The source width does explain a different discrepancy — the transverse footprint (Fig. 2c) — so the cylinder tier should move to a 1 mm source, or a true surface patch.
5. **MUAP scale and fibre geometry.** With the constant corrected and the grid centred on the muscle, the forearm pool’s MUAPs span 0.7–160 μV (median 24) and last 18 ms, inside the band of recorded surface MUAPs (tens to hundreds of microvolts, 5–20 ms), though the largest units stay below the 1–2 mV that the largest superficial units reach in recordings, and the amplitude carries the $\sigma_{\text{in}}\pi a^2$ caveat of Sec. 4. What remains is geometric: straight, full-length, constant-depth fibres give a surface field broader than a real unit’s, which the series-fiber experiment addresses. Amplitude cancellation inherits it (68% at 20% drive, where Keenan et al. [48] give 33%): 100 units firing 18 ms MUAPs from shared fibres overlap more than a real pool’s. We note for the record that an earlier grid layout in the repository (electrodes snapped to mesh vertices, its centre 17 mm off the muscle) combined with the uncorrected constant gave 0.6 μV and 33 ms, and was first read as a model property.
6. **Pool.** The Gaussian renewal process has no refractory floor (1.6% of intervals below 20 ms at 40 pps).
7. **Forearm mesh resolution.** The 42,315-cell forearm mesh has not been checked against a refined one; given that mesh structure in φ is what limits the recipe’s amplitude on finite-element fields (item 3), that study is due.
8. **No experimental confrontation yet.** The next validation step is spike-triggered surface MUAPs from decomposed HD-EMG on the same subject against the MRI pipeline, with the normalised error and correlation targets of Lowery et al. [13], Botelho et al. [14].

10 Conclusion and outlook

`emgforge` is a forward model whose every step can be tested, and has been. The synthesis recipe at its centre is the line-source integral done carefully — a monopole-fitted lead field, a second-derivative source, a one-sided tendon window and physical time — and it reproduces first principles exactly. The tests also found the model’s mistakes, and we have listed them rather than smoothed them over. The code, the validation suite and the datasets are public so that both the results and the open items can be checked.

Three uses are immediate. The interference-EMG trials of Sec. 8 carry ground-truth spike trains, unit properties and a full provenance record, which is what a decomposition benchmark such as MUniverse [17] consumes; an exporter to its BIDS-based layout is the natural next step. The lead fields of the parametric limb are the training data for neural-operator surrogates [20], and the validation tiers here give those surrogates a signal-level target (the SFAP, not only the field) to be held against. And a forward model that is exact against first principles is

the reference that model-informed decomposition [18] needs when it inverts a learned MUAP generator: the generator can be checked against this chain before it is inverted.

Code and data availability

Code: <https://github.com/dhalatsis/emgforge> (MIT), branch `paper/arxiv-v1`. The validation suite is `scripts/validation/run_all.py`; the figure and dataset scripts are in `paper/figures/`; dataset manifests in `paper/datasets/`.

Author contributions

D.H. designed and implemented the pipeline, the golden method, the activation layer, the validation suite and wrote the paper. N.E.-N. developed the harmonic-streamline fibre geometry. Contributions of P.M. and D.F. to be stated by the authors.

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A The validation checks

Table 5 lists every check of the suite with its criterion and the measured result, generated from the suite’s JSON records (`paper/make_checks_table.py`).

Table 5: Every check of the validation suite: what it asserts, its criterion (first clause), what was measured (first clause), and the outcome. Full strings are in `docs/validation/REPORT.md`.

check	criterion	measured	result
<i>Tier A — cylinder: first principles, analytical, FEM, pipeline</i>			
A0.1 spatial engine $\equiv$ first-principles line-source integral (shape, timing)	$r \geq 0.999$ and $ \text{lag} \leq 0.1$ ms: same integral, so only discretisation can differ	$r = [1.0, 0.9999, 0.9999]$, $\text{lag}(\text{ms}) = [-0.0, -0.05, -0.05]$ over NMJ offsets 0/−20/−30 mm	pass
A0.2 spatial engine amplitude constant equals first principles	ratio $1.00 \pm 5\%$, independent of v	engine/reference amplitude ratio = 0.994 at $v=2$, 0.994 at $v=4$	pass
A0.3 Fourier engine vs first principles (signed r , lag)	$r \geq +0.99$ (same integral, same sign, no lag)	signed $r = [-0.795, -0.684, -0.643]$ at lag $[1.4, 2.55, 0.85]$ ms; vs a spatially MIRRORED IAP: $r = [-0.788, -0.684, -0.642]$	known
A0.4a sampling invariance: fsamp 4096→2048 and dz 0.98→0.5 mm leave the SFAP unchanged	$r > 0.995$, amplitude within 2 % (discretisation-converged)	fsamp: $r=1.00000$ p2p ratio=0.9977; dz: $r=0.99886$ p2p ratio=1.0011	pass
A0.4b translation: moving the electrode Δz along the fibre delays the propagating lobe by $\Delta z/v$	$ \Delta \leq$ one sample (0.24 ms)	$\Delta z=20$ mm $\rightarrow$ main-lobe delay 4.88 ms (expect 5.00)	pass
A0.5 the CSD source is monopole-free: $\int i_m(z,t) dz = 0$ at every instant	$< 1e-6$: generation and end-of-fibre terms exactly balance the propagating tripoles	$\max_t \Sigma_z i_m / \Sigma_z i_m = 7.7e-17$ (boxcar), $7.7e-17$ (one_sided)	pass
A0.4c superposition: a two-fibre MUAP is the sum of its SFAPs	$< 1e-9$ (linear volume conductor)	$\max \text{MUAP} - (\text{SFAP} + \text{SFAP}) / \text{p2p} = 0.0e+00$	pass
A0.4d CV scaling: v 4→3 stretches the SFAP by 4/3 and scales its spectrum by 3/4	both within 8 %	duration ratio 1.328 (expect 1.333), MNF ratio 0.757 (expect 0.750)	pass
A1.1 FEM $\varphi(z)$ reproduces the analytical cylinder $\varphi(z)$ along the fibre (5 depths)	$r \geq 0.99$; FWHM within 20 % (electrode models differ: $\varnothing 10$ mm disk vs $\sigma=5$ mm Gaussian blob, so the shallowest fibres see...	$r = [0.9962, 0.9948, 0.9969, 0.9984, 0.9998]$; FWHM _{z} ana/FEM (mm) = [$41.0/35.1$, $53.1/45.3$, $63.1/57.7$, $68.9/65.8$, $81.2/82.1$] at $r = 33/30/27/25/20$ mm	pass
A1.1b ... and its second derivative $\varphi''(z)$ — the kernel the SFAP actually integrates	$r \geq 0.98$ at the best-matched electrode radius (the FEM electrode is a $\sigma=5$ mm Gaussian blob clipped by the skin — the analytical disk...	$r(\varphi'')$ vs analytical electrode radius 2.5/5/7.5/10 mm: $r=33$: $[0.985, 0.964, 0.916, 0.842]$; $r=27$: $[0.931, 0.923, 0.909, 0.889]$; $r=20$: $[1.0, 1.0, 1.0, \dots]$	fail
A1.2 FEM/analytical SFAP amplitude ratio is one constant across depth (same depth law)	TWO findings. (i) A smooth 1.3–1.4 $\times$ drift survives every preprocessing and a $\sigma=1$ mm source: the FEM cylinder decays $\sim 30\%$ slower...	p2p ratio FEM/ana, $r=33 \rightarrow 20$ mm, rel. to $r=33$ — golden: $[1.0, 1.46, 1.35, 1.2, 0.86]$ (spread 1.69 $\times$); no denoise: $[1.0, 1.59, 1.3, 1.72, 2.03]$...	pass

check	criterion	measured	result
A1.3 FEM $\varphi(z)$ vs analytical for fibres off the electrode meridian ($\theta = 10\text{--}45^\circ$)	$r \geq 0.99$	$r = [0.9978, 0.9997, 0.9985, 0.997]$	pass
A2.1 pipeline on FEM $\varphi \equiv$ pipeline on analytical φ (4 depths $\times$ 2 NMJ offsets)	$r \geq 0.95$ with the golden recipe; deep fibres ≥ 0.99 . The shallow-fibre residual is the FEM field (electrode model + mesh ripple through...	golden spatial: $r = [0.948, 0.97, 0.9, 0.929, 0.946, 0.915, 0.998, 0.995]$ (depth 33,33,30,30,25,25,20,20 mm; no denoise: $[0.948, 0.954, 0.891, 0.909, 0.919, \dots]$	known
A2.2 monopole denoise removes FEM mesh ripple without changing the analytical answer	denoised jaggedness $\leq$ raw; agreement not reduced	SFAP jaggedness = raw $\rightarrow$ denoised = $[0.004\rightarrow 0.003, 0.007\rightarrow 0.002, 0.006\rightarrow 0.003, 0.011\rightarrow 0.002]$	pass
A2.3 end-of-fibre onset lands at L/v in the FEM pipeline (and where the Farina generator puts it)	pipeline $ \Delta \leq 0.5$ ms	... expected/Farina/pipeline (ms): 10.0/7.57/10.26, 15.0/12.45/15.39 $\rightarrow$ pipeline $ \Delta \leq 0.39$ ms; Farina onset leads by up to 2.55 ms (its IAP body...	pass
A2.4 conduction velocity recovered from a 4-channel longitudinal array (both models)	within 5 % of the set CV, linear lag-distance $R^2 > 0.99$	Farina 3.83 m/s ($R^2=0.998$); FEM pipeline 3.98 m/s ($R^2=0.999$); set 4.0	pass
A2.5 waveform $ r $ vs the Farina generator (sign-agnostic, ± 6 ms lag)	$ r \geq 0.9$ — currently limited by the IAP-orientation/sign conventions of the Fourier/Farina family (A0.3)	spatial/Fourier: $-0.91/+0.94, -0.74/+0.74, -0.93/+0.52$ at NMJ 0/ $-20/-30$ mm	known
A3.1 rotational symmetry: electrode $+30^\circ \equiv$ fibre -30°	$r > 0.999$, peak within 3 % (mesh-discretisation level)	$r = [1.0, 1.0]$, peak ratio = $[0.9986, 1.0109]$	pass
A3.2 axial translation: electrode $+20$ mm $\equiv$ fibre window -20 mm (Neumann end effect)	$r > 0.995$, peak within 3 % — the residual (≤ 0.5 %) is the finite-length (240 mm) mesh's end effect	$r = [0.99865, 0.99561]$, peak ratio = $[0.9984, 0.9946]$ (electrode 100 mm from the mesh end)	pass
A3.3 reciprocity between two interior points in the anisotropic muscle	ratio 1 ± 5 % (symmetric $\sigma \Rightarrow$ Green's function symmetric)	$\varphi_A(B) = 2.3625e-02$, $\varphi_B(A) = 2.3311e-02$, ratio 1.0135	pass

Tier B — MUAP features against the literature

B1 monopolar SFAP over the fibre (between IZ and tendon) is triphasic $+ - +$ with the main lobe negative	$+ - +$ with the largest lobe negative (depolarised zone under the electrode). The Farina port's propagating main lobe is POSITIVE: its output...	pipeline lobes $[1, -1, 1]$ (largest -1); Farina generator lobes $[-1, 1, -1, 1, -1]$ (largest +1); electrode 30 mm from the NMJ, tendons ≥ 95 mm away,...	pass
B2 conduction velocity recovered from an SD array tracks the set CV (3, 4, 5 m/s)	within 5 % (pipeline) / 6 % (Farina) at each CV; physiological range 3–5 m/s	pipeline/Farina: $v=3$: 3.13/3.15 m/s, $v=4$: 3.98/4.20 m/s, $v=5$: 4.88/nan m/s (Farina NaN = its k-grid does not build at that v)	pass

check	criterion	measured	result
B3 innervation-zone signature: monopolar mirror symmetry, SD null and phase reversal at the IZ	mono $r > 0.99$; SD at the IZ $< 5\%$ of max; SD $r(+d, -d) < -0.95$ (mirror images of opposite sign)	pipeline: mono $r(+d, -d) = [0.9981, 0.9979, 0.9985]$, SD@IZ/max = 0.041, SD $r(+d, -d) = [-0.999, -0.998, -0.998]$; Farina: SD@IZ/max = 0.004, SD $r(+d, -d) = \dots$	pass
B4a the end-of-fibre component is non-propagating: same onset on every electrode	spread < 0.5 ms across electrodes $-30\dots+30$ mm	pipeline onsets [10.51, 10.51, 10.51, 10.51, 10.51] ms (spread 0.00); Farina [7.57, 7.81, 7.57, 7.57, 7.57] (spread 0.24); expected $L/v = 10$ ms	pass
B4b EOF / propagating amplitude ratio increases monotonically with depth	monotone increase (the propagating part decays faster than the far-field EOF)	ratio at depth-below-skin 7/10/13/15/20 mm = [0.021, 0.053, 0.097, 0.127, 0.204]	pass
B4c spatial filters suppress the EOF: EOF/propagating ratio mono $> SD > DD$	strict ordering	{'mono': 0.053, 'SD': 0.014, 'DD': 0.004} (IED 10 mm)	pass
B5 amplitude vs depth is a power law (log-log linear), steeper for SD than monopolar	$R^2 > 0.95$ over 7–25 mm; $n_SD > n_mono$; monotone decrease. Note the Farina generator's own exponent differs from the first-principles engine's...	exponent n ($p2p \propto d^{-n}$), R^2 : pipeline mono 2.94 (0.998), SD 3.48 (0.998); Farina generator mono 2.04 (0.986); FEM pipeline mono 3.17 (0.987)...	pass
B6 transverse spread: monopolar wider than SD (lit. 2.5–4 $\times$); depth $\approx 0.2 \times$ (mono 50...	mono $\geq SD$ width; both widen with depth. The literature ratio (2.5–4 $\times$; biceps MUs 15–25 mm deep, mono 72–96 mm vs SD 24–32 mm) comes...	50 %-width (mm) at depth 15/20 mm: mono 45/88, SD 41/53 (ratio 1.1/1.7); depth/(0.2-width) = 1.66/1.13 (Roeleveld's biceps-MU rule = 1.0; single fibre...	pass
B7 MUAP amplitude $\propto$ fibre count at fixed geometry; NMJ scatter ($\sigma=5$ mm) disperses and...	exact linearity; sub-linear amplitude and longer duration with dispersion	50 identical fibres = 50 $\times$ SFAP to 9.9e-16; with NMJ $\sigma=5$ mm: p2p 0.86 $\times$ the coherent sum, 10 %-duration 20.0 $\rightarrow$ 22.2 ms	pass
B8 the bipolar montage is a pure spatial difference of a translation-invariant wave ($\Rightarrow$ comb...	equivariance $< 5e-3$: $SD(t) = S(t) - S(t - IED/v)$ to that precision, so $ H = 2 \sin(\pi f \cdot IED/v) $ with nulls at $n \cdot v/IED$ follows...	translation equivariance residual (single travelling wave, 9–33 ms): no denoise 1.4e-03, golden 2.2e-03. Measured null of $ SD / mono $ — pipeline (no...	pass
B9 electrode size is a spatial low-pass: amplitude and MNF fall with electrode area; $\varnothing 5$ mm...	monotone decrease; -3 dB points as in the literature	Farina p2p vs radius (0.5, 2.5, 5.0, 10.0): [1.0, 0.98, 0.924, 0.765]; MNF: [81.7, 81.0, 78.8, 72.8] Hz; disc filter: $\varnothing 5$ mm @100 c/m = -2.8 dB,...	pass
B10 SD amplitude grows $\approx$ linearly with IED when IED λ , then saturates towards $\lambda/2$	differentiator regime then saturation ($\lambda \approx v \cdot T_lobe \approx 20\text{--}30$ mm)	SD p2p vs IED (2.5, 5.0, 10.0, 20.0) mm — pipeline: [1.0, 1.95, 3.59, 5.28]; Farina: [1.0, 1.95, 3.52, 5.01] ($\times 2$ IED $\rightarrow \geq 1.7\times$ at small...	pass
B11 subcutaneous fat attenuates (Kuiken 2003: $-31/-80/-90\%$ at 3/9/18 mm) and low-passes the surface...	within $\times 1.5$ of Kuiken's FE curve; RMS monotone decreasing and MNF lower at 3 and 9 mm than at 0.5 mm; FEM (smoothed φ) vs analytical...	analytical RMS retained at fat (0.5, 3.0, 6.0, 9.0, 18.0) mm: [1.0, 0.498, 0.312, 0.226, 0.124] (Kuiken 0.69/0.20/0.10 at 3/9/18); MNF [91.0, 79.0, 76.0, 79.0, ...	pass

check	criterion	measured	result
B12 muscle anisotropy ($\sigma_z/\sigma_r = 5$) elongates the lead field along the fibres	$\sqrt{5}$ exactly in the infinite medium; 1.3–3 in the layered cylinder (fat/skin layers and the bounded cross-section compress the elongation); FEM...	FWHM_z(aniso)/FWHM_z(iso) analytical cylinder 1.48, infinite medium 2.236 (theory $\sqrt{5} = 2.236$); FEM along/across FWHM at r=30: 2.29 (layered, bounded...	pass
<i>Tier C — interference EMG and motor-unit pool</i>			
C1 discharge rates lie in 5–40 pps and follow the onion skin (earlier-recruited fire faster)	model rates within 5–40 pps; measured within 10 % of nominal; $\rho(\text{threshold}, \text{rate}) < -0.9$	model: min rate 5, peak 40 pps; measured @drive 0.5: 89 active, 7.6–36.5 pps, Spearman(threshold, rate) = -0.98; @drive 1.0 max 42.9 pps (renewal-process...	pass
C2 inter-spike-interval variability: CoV 0.1–0.3, no ISI below the ~20 ms refractory floor	CoV [0.1, 0.3]; < 1 % of ISIs under 20 ms. The Gaussian renewal model has no refractory floor — at 40 pps (ISI 25 ms, σ 4 ms) sub-20 ms...	median per-MU ISI CoV 0.16 (model ISI_CV = 0.167); ISIs < 20 ms: 1.58 %; min ISI 9.8 ms	fail
C3 recruitment thresholds are right-skewed (many low, few high); twitch forces span $\approx 100\times$	skewness > 0.5; 50–200 $\times$ (Fuglevand RP = 100; FDI 130)	threshold skewness +1.25 (median/max = 0.12); twitch P_max/P_min = 95	pass
C4 amplitude PDF: super-Gaussian at low force, $\rightarrow$ Gaussian (kurtosis 3–3.5) at high; ARV/RMS [0.71,...	kurtosis decreasing with force to 2.8–3.6; ARV/RMS between Laplacian and Gaussian. NB: with the C-04 MUAPs (33 ms long) the low-force EMG is denser...	kurtosis @drive 0.05/0.1/0.5/0.9 = [5.59, 4.56, 3.1, 2.92]; ARV/RMS @0.3/0.7 = [0.768, 0.786] (Laplacian 0.707, Gaussian 0.798)	pass
C5 amplitude cancellation grows with excitation (≈ 30 % at 20 % $\rightarrow \approx 60$ % at maximum)	increasing; 15–60 % at 0.2, 40–85 % at 1.0 (Keenan 2005: 33 % $\rightarrow$ 62–65 %). Cancellation is set by MUAP duration $\times$ active-MU count:...	$1 - \frac{\text{mean} \Sigma \text{ trains} }{\Sigma \text{ mean} \text{train}_i }$ @drive 0.2/0.5/1.0 = [68, 79, 81] %	known
C6 EMG–force relation lies between linear and quadratic: RMS at 50 % force = 0.3–0.6 of RMS at 100 %	0.3 (quadratic, biceps/triceps) – 0.6 (linear, FDI/soleus); force monotone in drive. A ratio above 0.6 (EMG concave in force) follows from the...	RMS(50 % MVC)/RMS(100 % MVC) = 0.56; force plateau %MVC = [7, 16, 30, 46, 69, 85, 100] at drive [0.1, 0.2, 0.35, 0.5, 0.7, 0.85, 1.0]	pass
C7 interference-EMG spectrum: MNF/MDF at moderate force in 70–130 Hz	MDF 70–130 Hz (bipolar norms: biceps 90 ± 18 , TA 116 ± 20). Expected to fail while the C-04 fibre geometry yields 33 ms MUAPs — the spectrum...	@drive 0.5 (monopolar): MNF 81 Hz, MDF 77 Hz	pass
C8 larger motor units produce larger MUAPs (size amplitude on the recording side)	$\rho > 0.5$ (amplitude $\propto$ fibre count at fixed depth; depth scatter lowers ρ)	Spearman(fibres per MU, grid-max p2p) = +0.98 over 100 MUs (sizes 5–395 fibres)	pass
<i>Tier S — chain-level sanity</i>			
Orderly recruitment (Henneman size principle)	n_active non-decreasing in drive; smallest units recruited first	n_active @ drive [0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1.0] = [37, 51, 67, 84, 94, 100, 100]; thresholds sorted=True	pass

check	criterion	measured	result
Onion-skin rate coding	earlier (smaller) units discharge faster than later (larger) ones	@drive 0.7: first-recruited 40.0 Hz vs last 5.6 Hz (98 active)	pass
Interference-EMG amplitude $\uparrow$ with contraction	monotone increase with drive	plateau RMS(μ V) @ [0.1, 0.2, 0.35, 0.5, 0.7, 0.9] = [14.822, 28.279, 49.059, 61.24, 97.757, 124.795]	pass
EMG force: monotone and MVC-calibrated	monotone in drive; $\approx$ 100 %MVC at full drive	plateau force(%MVC) @ [0.1, 0.2, 0.35, 0.5, 0.7, 0.9, 1.0] = [7.4, 16.0, 30.3, 46.8, 69.8, 92.2, 102.5]	pass
Conduction velocity from HD-EMG propagation	physiological muscle-fibre CV, mean $\approx$ 4 m/s — the model uses one fibre CV ($\approx$ uniform across the pool)	pool median CV 4.32 m/s (range 4.1–4.8 over 100 MUs, IED 10.1 mm); corr(size, CV) = -0.41	pass
Spatial selectivity of a MUAP on the array	a MUAP is localised on the array, not a uniform far field (ratio > 1.5)	peak-channel RMS / grid-mean RMS = 2.16 (MU 99)	pass
Non-stationarity over a movement	EMG amplitude is modulated by joint angle, not just by the drive	RMS flexed / RMS extended = 8.66 (90.856 vs 10.495 μ V)	pass
MUAP physiological scale	single-MU surface MUAP $\approx$ 20–800 μ V, 5–20 ms (Merletti & Muceli 2019; Farina 2014) (the 0.6 μ V / 33 ms of earlier runs was an...	median p2p 23.60 μ V, median duration 18.1 ms	pass